MATH FOUNDATIONS

Statistics Visual Guide

Hypothesis tests, p-values, confidence intervals, and regression — inferring truth from samples.

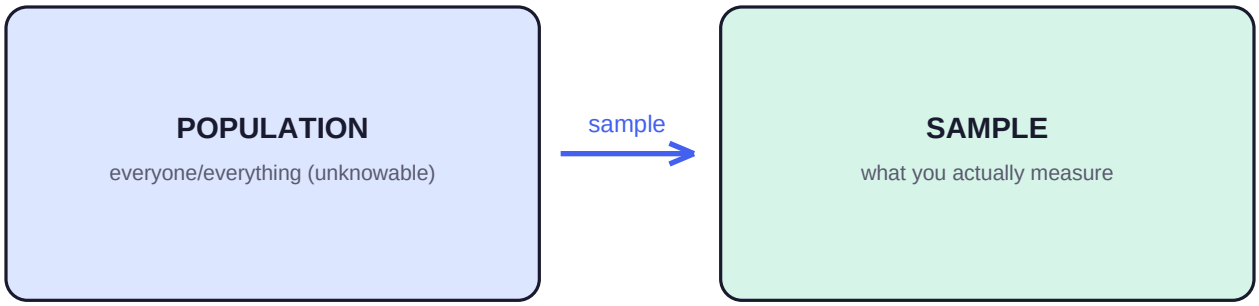
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Samples vs Populations

You measure a sample; you care about the population. The gap is the whole game.



BIAS

non-random sampling -> the sample lies about the population

No fancy test fixes a bad sample. Randomness of selection beats size of selection.

STEP 2/6

Confidence Intervals

A range that would trap the true value 95% of the time — if you repeated the study.

estimate +/- margin of error

e.g. 52% +/- 3% -> interval [49%, 55%]

WIDER when...

small n, high variance, high confidence

NARROWER when...

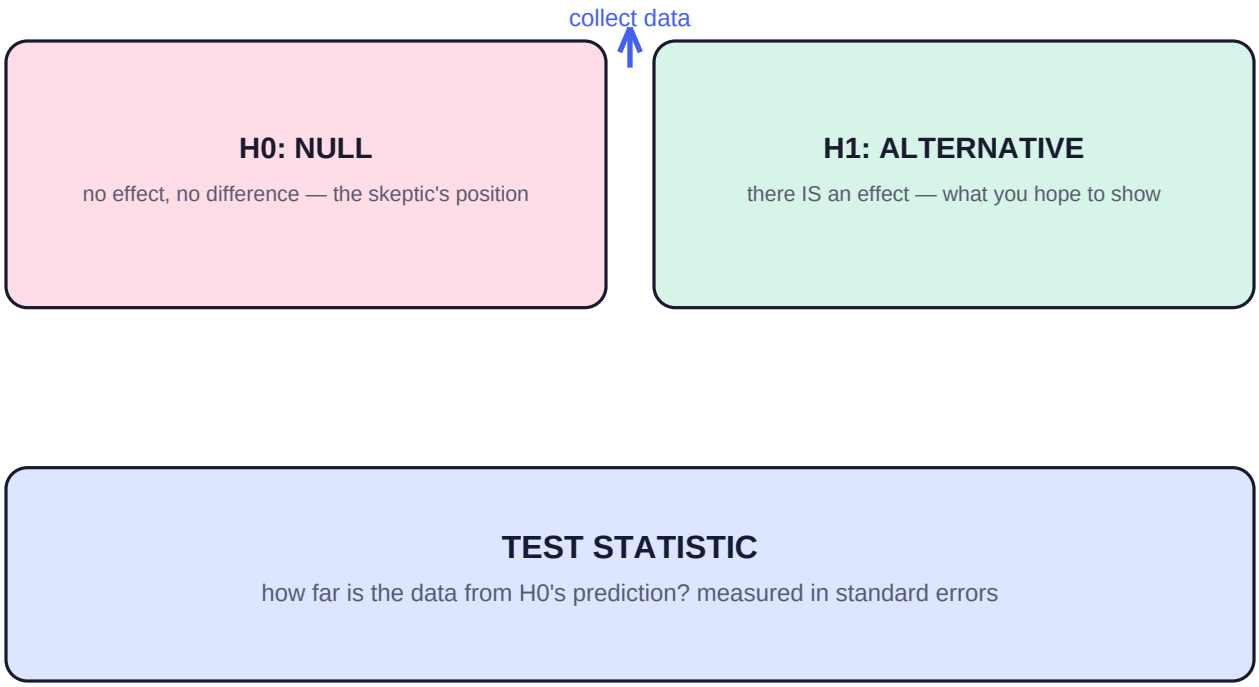
big n, low variance, lower confidence

A 95% CI is a property of the PROCEDURE, not a 95% chance the truth is inside this one interval.

STEP 3/6

Hypothesis Testing Framework

Assume nothing is happening (H0); ask if the data is surprising under that assumption.



You never 'prove' H1. You reject H0 — or fail to. Absence of evidence is not evidence of absence.

STEP 4/6

P-values and Error Types

What a p-value actually is — and the two ways to be wrong.

p = P(data this extreme | H0 true)

small p -> data surprises the skeptic -> reject H0

TYPE I (alpha)

reject a TRUE H0 — false alarm

TYPE II (beta)

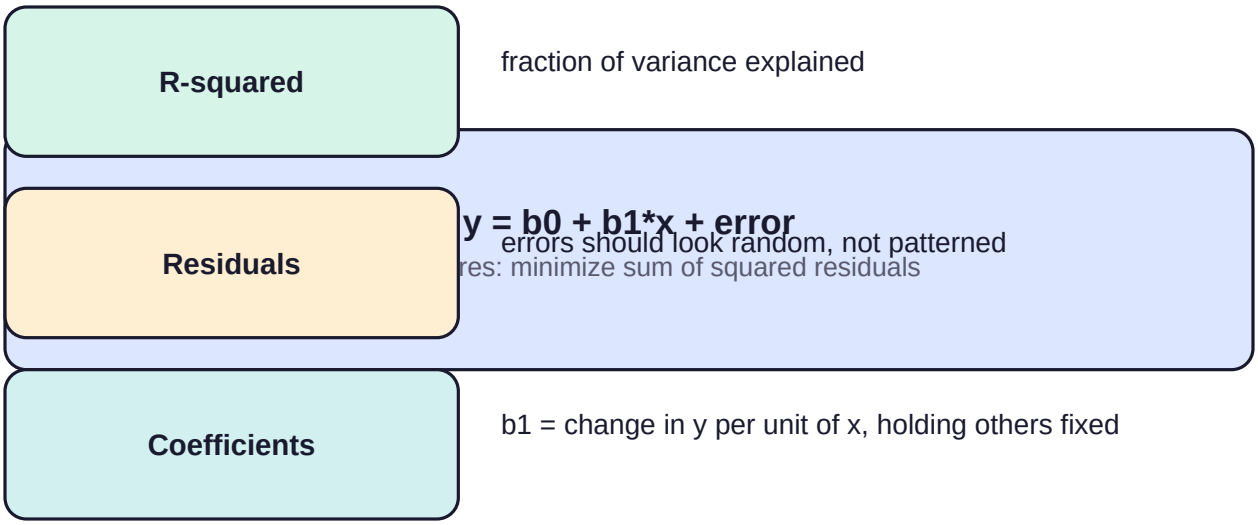
keep a FALSE H0 — missed detection

p < 0.05 is convention, not a law of nature. Report effect sizes, not just stars.

STEP 5/6

Regression Basics

Fit a line that minimizes squared error — then interrogate it.



STEP 6/6

Correlation, Causation & Other Traps

The statistics are easy. The interpretation is where people get hurt.

CORRELATION != CAUSATION

confounders, reverse causation, coincidence — always ask

MULTIPLE TESTING

20 tests at p<0.05 -> expect 1 false hit. Correct (Bonferroni/FDR).

P-HACKING

stopping when p dips, trying every slice — preregister analyses

STEP ✓/✓

Key Takeaways

Statistics — Remember

- You measure samples to learn about populations; biased sampling poisons everything downstream.
- Confidence intervals quantify uncertainty — a 95% CI is about the procedure, repeated.
- Hypothesis testing: assume H0, measure surprise. Reject or fail to reject — never 'prove'.
- A p-value is P(data | H0), not P(H0 | data). Mind Type I and Type II errors.
- Regression fits by least squares; check residuals and report effect sizes, not just significance.
- Correlation is not causation; correct for multiple testing; preregister to avoid p-hacking.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev